

Report

App Name: LingoBite

07.17.2025

*This **App Design Document Template** provides a structured way to plan, communicate, and document the essential artifacts of your team's app development process. It serves as a living blueprint that justifies all of the team's design decisions, ensures alignment among team members, and provides a source of information for future case studies.*

Each team is responsible for maintaining a well-structured design document throughout the course. You may tailor the document as you see fit, as long as required sections from course assignments are included. The first few sections have been provided for you.

Team Member Names:

- Catherine Tang
- Eden Huang
- Mihi Nagariya
- Samia Amin

App Description:

- Menu translation app designed to help people understand restaurant menus written in foreign languages

App Name ideas:

- DishSelfie
- DishCam !
- LingoBite ! !(!?) (this is the one for now)
- Lingua
- FoodLingo !
- Menufy !! (SADLY TAKEN)
- The Menu (App)
- Menmoo !!
- Me&moo
- Menu Flex
- Mexu
- Bento - The Menu App !

Understand (Empathize, Define)

[Interview Plan](#)

Needs Statement:

Jane is an international worker who loves to travel. She is deciding what to eat in a foreign country while keeping in mind her allergies and food preferences. Her goal is to try out new dishes that she'll enjoy and isn't harmful to her.

[Empathy Map](#)

Explore (Ideate, Prototype)

Materialize (Usability/Concept Testing)

Interview Plan

Cognitive Science 112: Uncovering the Cognitive Science Behind the User Experience

INTERVIEW PLAN

A plan helps you define *why* you're conducting the interview and *what* you hope to learn. This clarity ensures your questions stay focused and relevant. Planning provides a logical flow of questions—from warm-up to deeper inquiries—so the conversation feels natural and productive, not scattered or awkward. Have a conversation with your group and ensure everyone is on the same page by answering the questions below.

1. Who do we want to learn from (i.e. who do we think is our target user)?

- People who enjoy trying different cuisines
- People traveling to countries where a different language is spoken
- People who want to make well-informed decisions when ordering from foreign menus that lack photos or have only brief descriptions

2. What do we want to learn from them? What problem do we believe they have that we can solve?

- We want to learn what customers are looking for in a translation app and why they would need it. What are the most common activities when traveling (e.g., restaurants, hotels, etc.)
- We want to understand their experience when entering a restaurant and having difficulty understanding the menu.
- We believe we can help customers feel more welcomed and comfortable by providing accurate translations and food images to help them overcome this obstacle.

3. How will we find potential users to interview?

- International students currently living in or attending UC Berkeley over the summer who are reading English menus while eating out
 - Many international students in my class are not fluent in English
- People whose first language isn't English
- Food lovers who enjoy trying new cuisines
- People who like to eat out but are shy to ask about menu items

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4. What questions will we ask to find out if we're right or not? List between 3 - 7 questions for your interview.

- Date/time of interview:
 - General Description of interviewee:
1. Can you tell me about the last time you had to order from a menu but didn't fully understand the options?
 2. What did you do in that situation?
 3. What made it difficult to understand the menu?
 4. How did you decide what to order in the end?
 5. Was there anything you liked or disliked about that experience?
 6. What made you look up a picture of a dish online before ordering?
 7. How long did it take you when ordering a menu item in another language?

5. What is your script to introduce yourself, the group and what you're trying to accomplish? Remember, this is not the time to talk about your solution.

Ex. Thanks for taking time to talk with me today. My name is _____ and I am taking a course on user experience design. My team and I are conducting interviews to learn more about how people find childcare in a new city.Our conversation is confidential and will only be shared with my team mates, my teacher and other students within the class...

Thank you for taking the time to talk with me today, I really appreciate it. My name is _____ and I'm taking a course on user experience design. My team and I are trying to understand how people navigate through menus that they do not understand, and the difficulties that they experience. Our conversation will be confidential and will only be shared with my teammates, teacher, and other students within the class.

6. When will you meet as a group to discuss your findings and build an Empathy Map?

We will meet on Wednesday to identify common patterns and observations from our interviews and to build our empathy map.

Interview Notes

INTERVIEW NOTES

Each team member will share notes from each of their interviews here so the rest of the team may review them and gain insights. Create a separate section for each team member. Be sure to include information about the interview: date, time, general description of interviewee (e.g. 27-year old single father and architect), questions from the team's interview plan and documented responses.

[\[Team member name\]'s Interview Notes](#)

[Interview #1](#)

[Interview #2](#)

[\[Team member name\]'s Interview Notes](#)

[Interview #1](#)

[Interview #2](#)

[\[Team member name\]'s Interview Notes](#)

[Interview #1](#)

[Interview #2](#)

[\[Team member name\]'s Interview Notes](#)

[Interview #1](#)

[Interview #2](#)

[Catherine]'s Interview Notes

Interview #1

Date: Jul 28

Time: 2:50 pm

Interviewee Description: 20 years old college student, a food lover who likes to travel

Documented Questions & Answers:

1. Can you tell me about the last time you had to order from a menu but didn't fully understand the options?

It happened several times, at Chinese restaurants. I couldn't understand things super well because the menu was in Chinese.

2. What did you do in that situation?

I took a photo of the menu and inputted it into Google Translate.

3. What made it difficult to understand the menu?

What made understanding the menu difficult were the language barrier, no pictures, no descriptions and hard to make inferences, sometimes it is hard to tell what the ingredients are even when there are pictures provided, and only the pictures of the most popular food are offered so it was hard to make well informed choices.

4. How did you decide what to order in the end?

I ordered popular dishes and dishes I am familiar with.

Follow-up: Were you satisfied with the choices?

I was okay, I wish I could have ordered new things but I was just not sure of the dish and their ingredients.

5. Was there anything you liked or disliked about that experience?

I liked learning about new dishes but wish I found out more in detail.

[Optional follow-up; only if the interviewee mentions searching for images of dishes online as a tactic to decide what to eat]

- **Have you ever looked up a picture of a dish online before ordering? What made you do that?**

Yes, I did that so that I can have a closer look at the ingredients.

6. How long did it take you when ordering a menu item in another language?

Hard to say, I am guessing 10-15 min, I am very picky.

Interview #2

Date: Jul 28

Time: 9pm

Interviewee Description: 20 years old college student who has high standards for food

Documented Questions & Answers:

- 1. Can you tell me about the last time you had to order from a menu but didn't fully understand the options?**

The menu labels were in Italian because I was abroad.

- 2. What did you do in that situation?**

Used context clues from words that I already knew.

- 3. What made it difficult to understand the menu?**

There were no pictures, and I did not fully understand things.

- 4. How did you decide what to order in the end?**

I went with my instincts after looking on social media.

Follow-up: Were you satisfied with the choices ?

Somewhat, I did not put in all the effort

- 5. Was there anything you liked or disliked about that experience?**

If there were more pictures and other resources I would have ordered something more to my preference, but I was just not feeling like putting in the energy into making decisions.

[Optional follow-up; only if the interviewee mentions searching for images of dishes online as a tactic to decide what to eat]

- Have you ever looked up a picture of a dish online before ordering? What made you do that?**

Yes, because the menu is in a different language.

- 6. How long did it take you when ordering a menu item in another language?**

Server came by 3 times, around 25 minutes

Samia's Interview Notes

Interview #1

Date: July 28, 2025

Time: 6:30pm

Interviewee Description: Family member, loves to eat at restaurants

Documented Questions & Answers:

1. Can you tell me about the last time you had to order from a menu but didn't fully understand the options?

Was at a small restaurant in Japan, could not understand the menu at all as we were in the small town and not a big city like Tokyo where people had English menus for tourists.

2. What did you do in that situation?

I asked the person next to me to translate as they happen to speak both Japanese and English.

3. What made it difficult to understand the menu?

The menu had native ingredients and meals native to Japan and was hard to find anywhere else.

4. How did you decide what to order in the end?

I got the same thing the person next to me ordered as they were a Japan native and knew the dishes best. I had one of the best meals ever and did not regret making that decision.

5. Was there anything you liked or disliked about that experience?

I liked how the people around me were very friendly and helpful.

Interview #2

Date: Jul 29, 2025

Time: 8:30am

Interviewee Description: Uncle, is a chef

Documented Questions & Answers:

1. Can you tell me about the last time you had to order from a menu but didn't fully understand the options?

I remember dining at a restaurant in Greece where the menu was entirely in Greek with no English translations and I couldn't understand most of the dishes.

2. What did you do in that situation?

The menu had pictures so I went by context clues and what dishes look best.

3. What made it difficult to understand the menu?

The fact that it was in a different language and the pictures were misleading. They also had different fishes that I did not know about.

4. How did you decide what to order in the end?

I asked the waiter who had broken English what was the most popular dish and I went with that.

5. Was there anything you liked or disliked about that experience?

I liked the fact that I got to try something new but I disliked how difficult it was to order food in a new country.

Eden's Interview Notes

Interview #1

Date: July 28th, 2025

Time: 11:55AM

Interviewee Description: 20-year old Male Korean International Student

Documented Questions & Answers:

Q: Can you tell me about the last time you had to order from a menu but didn't fully understand the options?

A: Yesterday maybe, I can't really remember. Different from the Korean menu, though everything went alright ok.

Q: What did you do in that situation?

A: I searched the internet or asked about AI or chatGPT. I usually eat at the international house, if there are unfamiliar options, I look them up or ask chatGPT.

Q: What made it difficult to understand the menu?

A: Unfamiliar with the menu (different from Korean menu's), new ingredients, and there are so many options, such as vegan, even with just milk, there were so many types of milk it made it hard to understand. I also usually eat at burger places or dorms so not much experience with eating out.

Q: How did you decide what to order in the end?

A: I look at the menu item on the internet for pictures and descriptions. But I often still end up choosing the wrong thing. Oh and I also look at the reviews.

Q: Was there anything you liked or disliked about that experience?

A: It was a bit inconvenient to look at the menu item I didn't know, but it was still helpful in looking at the pictures and reviews before deciding. Takes time, every time searching is inconvenient.

Q: What made you look up a picture of a dish online before ordering?

A: I thought it would be nice, if there were explanations of the food, especially for international user. I didn't want to make the wrong choice so I was searching for reviews on the menu items. For example, in the internal house, I especially cannot eat cilantro and there are so many dishes that had it. So I am always afraid

Q: How long did it take you when ordering a menu item in another language?

A: five or ten minutes before I order from the menu so I always have to search. When I went with my friend it took a long time, deciding whether items were vegetarian or have cilantro, not good at english so we had to translate and search. And the menus

are unfamiliar. My friends check vegetarian and I check cilantro. It would be a convenient app to translate and provide information.

Interview #2

Date: July 28th, 2025

Time: 6:17PM

Interviewee Description: 19-year old Female Chinese International Student

Documented Questions & Answers:

Q: Can you tell me about the last time you had to order from a menu but didn't fully understand the options?

A: You asked the right person. Today, I went the restaurant in restaurant menu, many unfamiliar words, very difficult for us to recognize the dishes, so we used google to search the restaurant the pictures and show pictures we want to the waiters to order food.

Q: What did you do in that situation?

A: searched google restaurant on the app and the reviews and picture of the item they wanted on the pictures

Q: What made it difficult to understand the menu?

A: The name of the dishes, very unfamiliar with some food names, even though I used the app to translate to my language it still took a lot of time to go through the menu and find what I wanted.

Q: How did you decide what to order in the end?

A: It depends on the pictures I searched on google and its price.

Q: Was there anything you liked or disliked about that experience?

A: Disliked that the name of food on the menu is not highly related to the things they serve.

Q: What made you look up a picture of a dish online before ordering?

A: Feedback about the food because the menu didn't show a picture on the menu, just a long name and I don't know what they are. I can only recognize the food with pictures.

Q: How long did it take you when ordering a menu item in another language?

A: Maybe, fifteen to twenty minutes because we had three people.

[Mihi]'s Interview Notes

Interview #1

Date: July 28th

Time: 8:56 P.M.

Interviewee Description: 22-year-old who loves to try new cuisines

Documented Questions & Answers:

1. Can you tell me about the last time you had to order from a menu but didn't fully understand?

- last week, when I tried Udon for the last time
- menu was in English, but didn't quite understand what the dishes were because I'd never have Udon before so it was new to me

2. What did you do in that situation?

- Asked friends what was good and dishes I thought looked interesting
- Searched up the name of the dish and looked at the images
- Asked ChatGPT (ex question: tell me what this dish is)

3. What made it difficult to understand the menu?

- I'd never been to an Udon restaurant before
- Didn't understand the words they were using even though they were in English (ex: didn't understand what "udon" was)
- Emphasized that it was mostly because of the lingo, although it was nice to learn

4. How did you decide what to order in the end?

- Combination of asking my friends + ChatGPT + images of the dish

5. Was there anything you liked or disliked about that experience?

- Not really, it would have been nice if the menu had pictures

6. Have you ever looked up a picture of a dish online before ordering? What made you do that?

- Yup, I just wanted to see what the dish looked like or didn't understand what the dish was

7. How long did it take you when ordering a menu item in a different language?

- Even though I've tried new cuisines, the menu has been in English

Interview #2

Date: July 29th

Time: 4:43 P.M.

Interviewee Description: 75-year-old whose first language is Hindi

Documented Questions & Answers:

1. Can you tell me about the last time you had to order from a menu but didn't fully understand?

- It's happened frequently
- I couldn't understand it because of the names of the dishes

2. What did you do in that situation?

- My grandchildren tell me what the dish consists of
- I ask the waiter what kind of dish it is
 - Ex: the flavor, sweet or spicy, if it's vegetarian

3. What made it difficult to understand the menu?

- I don't understand the names of the dishes
 - ex: words → "barbecue", "sizzling"

4. How did you decide what to order in the end?

- I look at the vegetarian options and look at the ingredients I am familiar with

5. Was there anything you liked or disliked about that experience?

- Sometimes when I don't understand what to order, I just make do with whatever there is
- I like it when the dishes I like and I am familiar with are on the menu

6. How long did it take you when ordering a menu item in a different language?

- A long time, I can't really do it by myself in a different language
- My grandkids help me out

Empathy Map

What are we trying to achieve?

What are we trying to achieve?

What are we trying to achieve?

What are we trying to achieve?

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What are we trying to achieve?

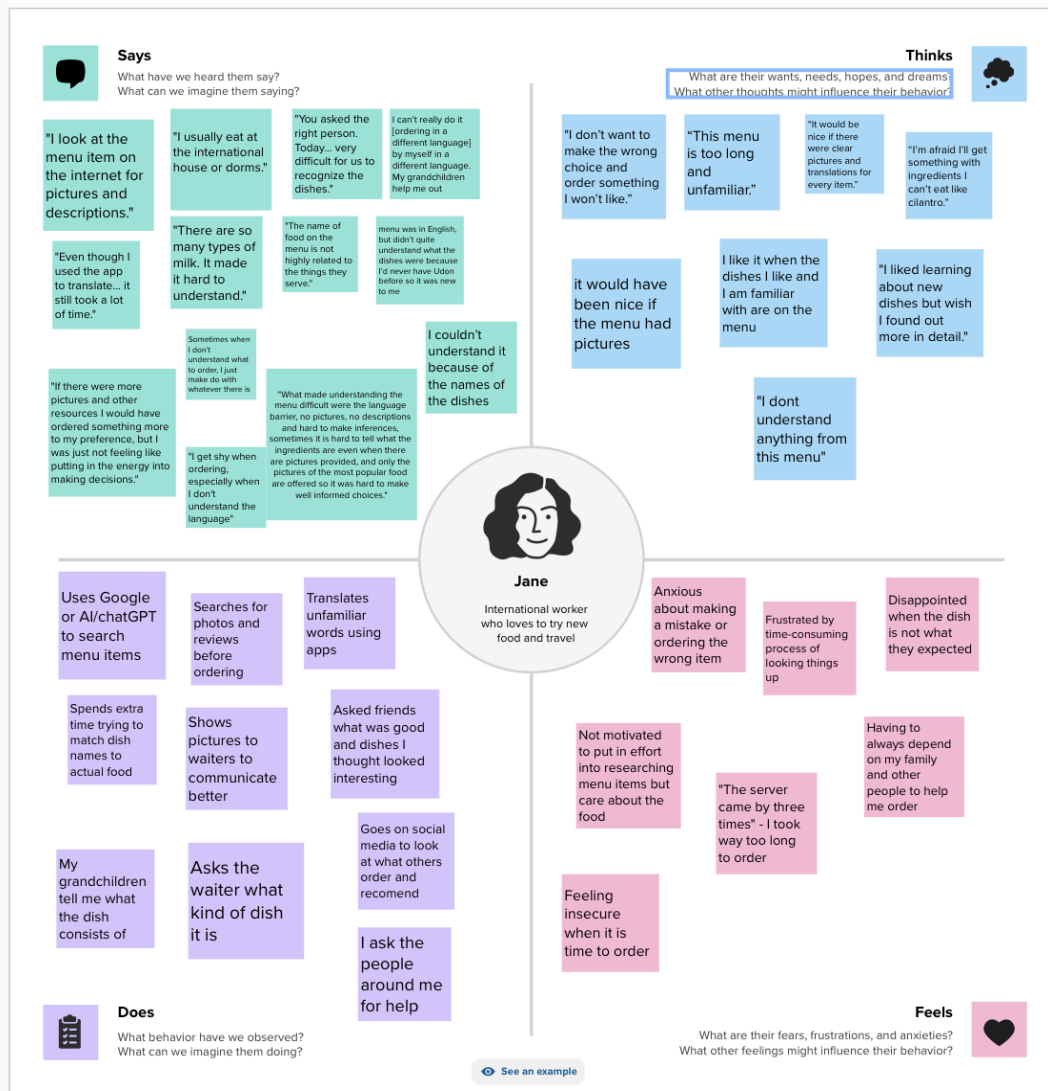
What are we trying to achieve?

What are we trying to achieve?

What are we trying to achieve?

Needs Statement:

Jane is an international worker who loves to travel. She is deciding what to eat in a foreign country while keeping in mind her allergies and food preferences. Her goal is to try out new dishes that she'll enjoy and isn't harmful to her.



Journey Map

Journey Map

Use this area to notestorm the steps in the your users' process (i.e. what are the steps they will need to complete to accomplish their goal using your app?). Write these steps down, discuss and rearrange them until you have a refined process that will go into your journey map below.

Jane is an international worker who loves to travel. She is deciding what to eat in a foreign country while keeping in mind her allergies and food preferences. Her goal is to try out new dishes that she'll enjoy and isn't harmful to her.

Jane lands in Korea

Jane sits down at a restaurant and looks at the menu

She doesn't understand the language on the menu

downloads the app

The app offers a tutorial that introduces her to the main features

Clicks on the camera feature and takes a picture of the menu

Translate Menu and Generating Pictures of Menu Options

Finish Deciding What to Order

Translate Order Name back to original Language For Order

Gets her food!

Search menu and U.S. menu are different

Jane asks around for help

Rate the Restaurant

Take Picture of Food and Write a review

Scenario: Solo Adult travel experience eating at an Foreign Restaurant without Translation Options



Jane

Steps

- 1 Jane sits down at a restaurant and looks at the menu
- 2 She doesn't understand the language on the menu
- 3 downloads the app
- 4 The app offers a tutorial that introduces her to the main features
- 5 Clicks on the camera feature and takes a picture of the menu
- 6 Translate Menu and Generating Pictures of Menu Options
- 7 Finish Deciding What to Order
- 8 Translate Order Name back to original Language to Order
- 9 Take Picture of Food
- 10 Write a Review



Doing

What is the user actively doing at each step?
What tools or resources is the user utilizing?
Are there any actions the user is avoiding?

- Looking at other customers food
- Looking for finding clues
- Searches the name of the app on the App Store
- Learning about the main features and how to use them
- looks at the rating and reviews of the app on App Store
- actively looking for the translation feature
- researching the menu items
- waiting for and looking at the translated menu items, example images
- Goes back and forth between the images of her top options
- Looks for pronunciations or points to what they want
- uses her camera to snap a picture of her food
- upload pictures and comments
- Clicks download and waits for it to finish downloading
- Compares the price difference



Thinking

What's on the user's mind at each step?
What doubts or concerns does the user have?
What decisions is the user trying to make?

- "I wonder what smells so good?"
- "Where can I find more information about these Menu options?"
- "will this app help?"
- "How much info will the app ask before I can get the translations?"
- "I hope the camera will capture everything!"
- "Are these images show what I will be really ordering?"
- "Should I order this dish or this one?"
- "Now I can order the dish of my choice!"
- "That was delicious"
- "I'm glad I was able to step out of my comfort zone."
- "I am starving!"
- "I'm too afraid to ask someone for help."
- "Hopefully this is easy to use"
- "so happy I was able to try out a new cuisine"



Feeling

What emotions is the user experiencing in each step?
How does the user's emotional state affect their actions?
Are these feelings positive, negative, or neutral?

- Curiosity about other customers food
- Frustration and Confusion of Menu Options
- satisfied if the app is quick and easy to install
- Impatience filling out information and going through tutorial anticipation for what comes next
- anticipation for translations
- satisfied and informed, or more frustrated
- Disappointment if none of the menu option as to the users liking
- determined to find a dish they like
- excited to use this app more
- Content with the food and service
- Satisfied with what the app offers
- Awkward and Discomfort eating someone in public or in a formal setting
- Impatience if the download takes too long
- anticipation for what comes next
- satisfied if menu item ingredients and images is to the users liking
- Proud of herself for using the app to help her order a dish in a new place



Opportunities

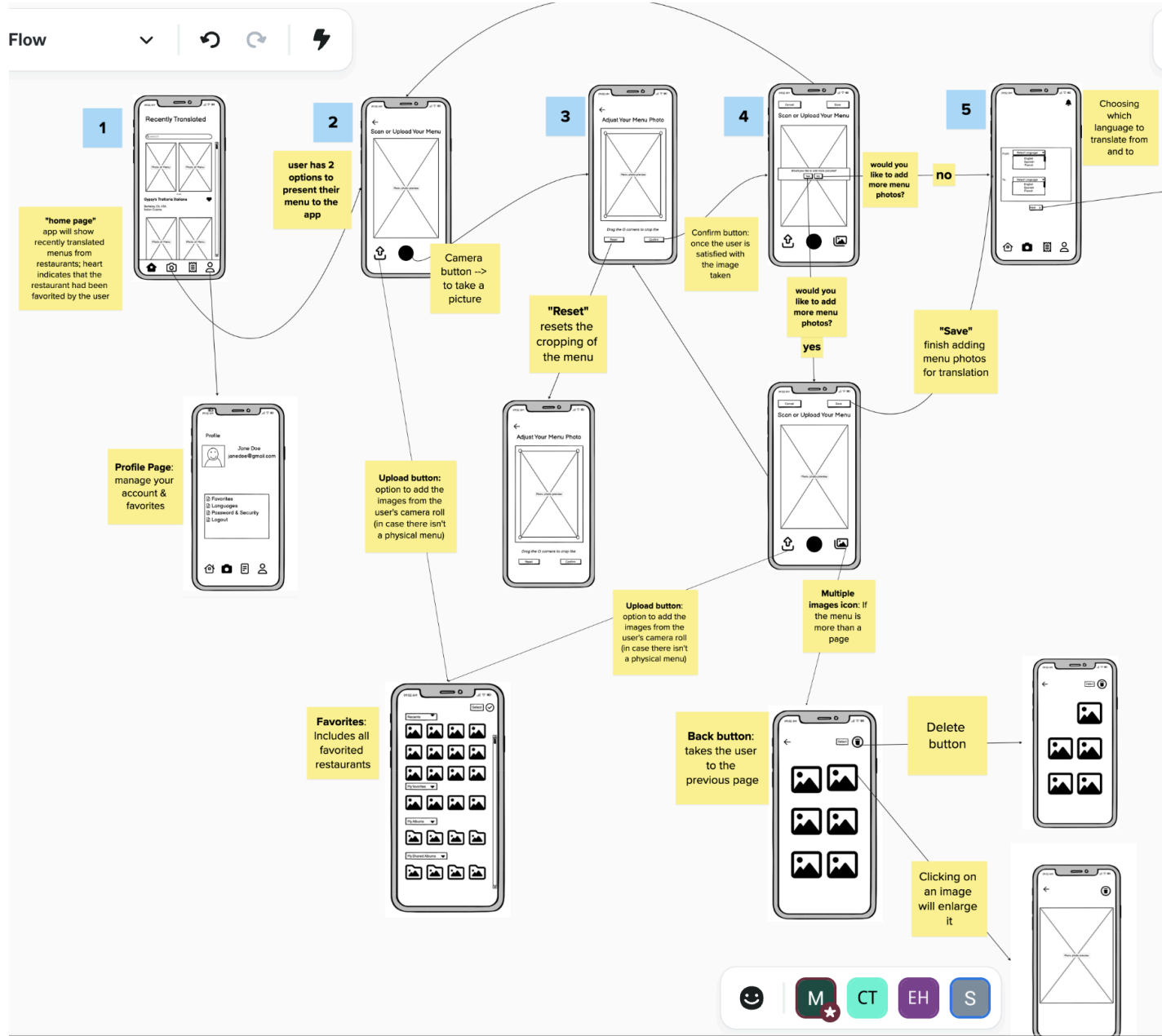
How might we mitigate some of the potential pain points?

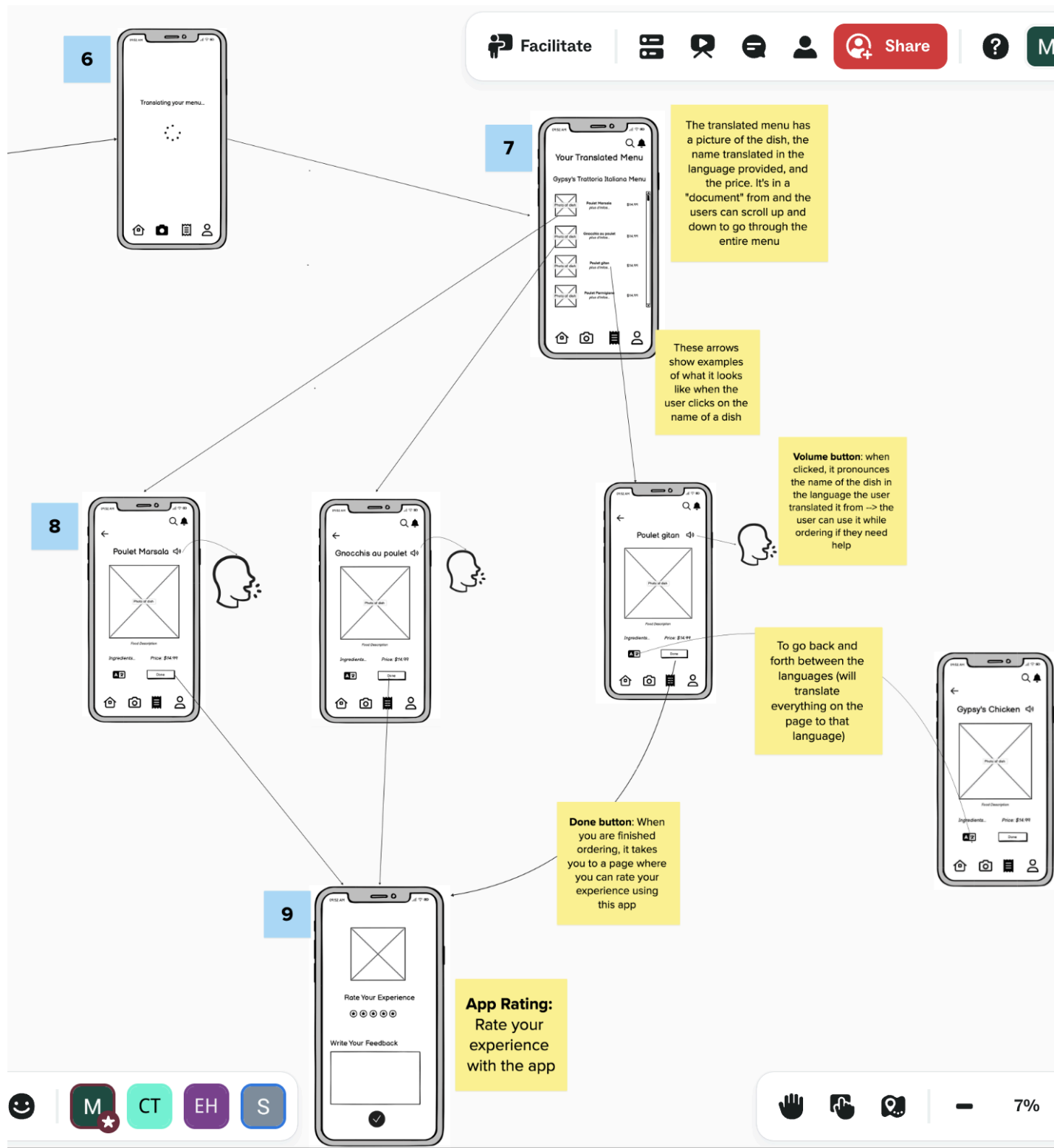
- app not in the consideration of the user yet
- having the right keywords for the user to be able to spot the app quickly
- Already having the app downloaded to the user doesn't have to deal with the confusion or frustration
- Warning the user with a message that helps them feel at ease and decreases their anxiety
- A symbol of some sort to show how much longer until the app translates
- The image should be retrieved from reliable platforms (ie. Doordash)
- response time should be fast so the user doesn't waste a lot of time waiting while trying to click on different dishes
- Provide the name of the dish in original language so that it is easy to order from the original menu
- upload time for the image shouldn't take that long
- easy access to all the comments
- The Country's Language of Translation is not available on the app yet
- restaurant in remote areas; no wifi
- An "x" option on the top right corner so the user can exit out of the tutorial anytime they like
- Search button to search the name of the restaurant
- See an example

UI Flow

Mural Link:

<https://app.mural.co/t/cogsci1121830/m/cogsci1121830/1754247177661/c45b7173de5cb2116582077196505f1adb1e469a?sender=u08115277f420ecde4c7b1599>





Peer Critique

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Peer Critique

Provide objective and constructive feedback to your classmates by answering the questions below. Give examples to support your comments by referencing specific frames from the design.

Peer Reviewer #1

Name: Annabelle Wu

Discoverability. Is it possible to determine what actions are possible and the current state of the device?
Comments: Yes, all the actions on each screen are very clearly shown with corresponding icons or text labels. All the icons and the placement of action items are intuitive to users as they match pre-existing mental models of similar functions (i.e. camera screen with shutter and upload functions, bottom bar leading to different pages). The state of the device is also easily determinable from each interface.
Feedback. Is it easy to determine the current or new state after an action has been completed?
Comments: Yes, this flow incorporates lots of feedback components that make it very clear the state after completing a certain action. These include pop-up windows after uploading images asking if the user would like to upload more, images enlarging when you click on them from an album, and images disappearing after being deleted.

Peer Reviewer #2

Name: Ashley Leng

Affordances. Do affordances exist that make the desired actions possible?
Comments: Yes, there are multiple affordances present in the design. The circular camera button affords pressing, which allows for taking photos and creating a section for user input

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The four icons on the bottom row afford pressing, and their constant presence in most of the screens allows for easy navigation. The scroll bar in the favorites section affords scrolling, which affords extra storage. Users know just by glancing at the scroll bar that they are allotted more space. The (mostly) consistent presence of back buttons affords room for error and easy navigation. However, certain screens (like the profile page) don't have a back button. Users can exit by pressing one of the four icons on the bottom, but understanding that might take a second. Adding more backbuttons could potentially make the design more intuitive.

Signifiers. Are signifiers used? Do they promote discoverability and ensure the feedback is well communicated and intelligible?

Comments: Yes, there are many signifiers throughout the design. The symbols of the icons on the bottom signify their corresponding usages, i.e. the camera signifies taking photos and the house icon signifies a home screen / menu. These are commonly used icons, making learning how to use the application easy for the users. Some screens have text that concisely explains to users what to do. Since the text is so short and simple, I don't think it detracts from the design. However, the delete function is a bit confusing because there's no signifier for how to delete (ex: small x icons in the corner of photos) and there doesn't seem to be enough feedback. What happens after users press the delete icon? Do they simply press an image they want to delete after? How do they exit the delete function?

Peer Reviewer #3

Name: Jong Young Lee

Mappings. Is the relationship between controls and their actions easy to learn and use? Are the mappings enhanced as much as possible through spatial layout and temporal contiguity?

Comments: Yes, the mappings are well planned out. The relationship between controls of the buttons and their actions are easy to learn and use. The icons on the menu bar are well mapped to their functions, which enhance usability. There is a clear sequential flow with numbered layout from 1 to 9. This helps the users to understand the step by step process better. Each stage of mapping flows logically from the previous one to the next. There is good temporal sequencing with image selection to viewing. Key controls are placed in familiar locations at the bottom, reinforcing expectations and reducing the learning curve with a consistent spatial layout.

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Constraints. Does the UI provide physical, logical, semantic, and cultural constraints that guide actions and ease interpretation?

Comments: Yes, the constraints work well. There are limited options of buttons that a user could access. The options to either select done, go back, or delete are well provided. They guide actions well and their intentions are well interpreted. Limiting photo upload to camera or gallery prevents the user errors and aligns with the use case. Users also must decide on a language before translating, which could prevent errors. The guidance through prompts like "would you like to add more pictures?" prevents ambiguity and clarifies the intent of the next step. An addition of indicators for invalid images could help out the process as blurry photos or unsupported files could lead to failures. Moreover, having a manual input option could be a good addition as there is over-reliance on image input. There could be issues with recognizing the text if the photos have poor lighting or are blurry.

Peer Reviewer #4

Name: Vinaya Sivakumar

Conceptual model. Does the design communicate all the information needed to understand the system and how to operate it?

Comments: Yes, the design is clear and effectively communicates all the necessary information to understand and operate the system. The UI flow intuitively guides users through key tasks such as uploading a menu, translating it into a different language, and viewing or uploading dishes from a specific restaurant. The layout, iconography, and clearly defined calls to action contribute to a user-friendly and easy-to-navigate experience.

To further strengthen the design, it may be helpful to include a few static pages that briefly explain the app's purpose and highlight its core features at a glance. This addition could provide users with a better understanding of the app's value proposition from the outset.

Overall, the design does a great job of presenting complex functionality in an intuitive and accessible way.

Peer Reviewer #5

Name: Danica Hartawan

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Discoverability: Is it possible to determine what actions are possible and the current state of the device?

Comments: First of all! This looks so amazing Catherine, Eden, Mihi and Samia! I think the team did a great job making the purpose of the app clear through both layout and icon placement. It's very obvious that this is an app to translate a MENU, unlike other tools in the market, and this app made that very discoverable. As someone who is bilingual and translating constantly for family that visits (who only speak Indonesian), I'm always on the lookout for apps that solve a very specific need, in this case translating restaurant menus. From the UI flow, I could immediately tell this was a tool for scanning, translating and editing food menus, important as Google Translate itself doesn't get the specifics of what I need. The presence of icons (like the camera and document outline) help users immediately recognize "Okay, this is for scanning". The step-by-step transitions, from photo -> crop -> confirm -> translate, make it very intuitive what you're allowed to do on each screen. The yellow sticky notes throughout also help explain each action without overloading visual space that often overstimulates readers like me.

In terms of state, It's also clear when you're editing versus uploading or confirming. It doesn't feel overwhelming, which is great for travelers or non-technical users like my parents or grandparents.

Feedback. Is it easy to determine the current or new state after an action has been completed?

Comments: Yes, the team did a great job showing clear state transitions between screens. For example, when the user clicks "upload" or "save", the app clearly responds with a new screen and options, like choosing between translations or saving to favourites. I think this feedback is especially important for non-native English speakers as it does reduce ambiguity.

One thing I personally appreciated is the visual indicator of cropping or resetting the menu image, it reassures the user that yes, their actions are being tracked and not lost. Also, it's a nice touch seeing the final translation overlaid onto the menu gives that satisfying feedback.

My only suggestion would be to explore whether there's a persistent indicator or micro-feedback element (like a checkmark or animation) that shows something was saved or favorited. Right now, the feedback is mostly through screen change, which works well, but small micro interactions could make it even more intuitive!



**Cognitive Science 112: Uncovering the Cognitive Science
Behind the User Experience**

User Test

1) Script:

Hello __, thank you for joining me/us today. We'd like to get your feedback on our App design. It's a menu translation app designed to help people understand restaurant menus written in foreign languages. I'm going to give you a task and please complete it by clicking through the screens. Additionally, please think out loud as you are going through the task.

2) [Think- Aloud Testing](#)

3) [Concept Testing](#)

4) Interview Notes

- a) [Eden's Interview Notes](#)
- b) [Mihi's Interview Notes](#)
- c) [Samia's Interview Notes](#)
- d) [Catherine's Interview Notes](#)

5) Write a summary of your findings (200 - 400 words).

Word count: 399

We think that our users will find our app intuitive and effective. Starting from the home page, the user would first notice the camera icon and search bar. The camera page and rating page should be easy to navigate. Some people could find the cropping page unnecessary or helpful, depending on their situation. The (French) translated menu would be familiar to people who use delivery apps, but may be difficult without categorical outlines, and the detailed dish pages may be overloaded with information.

Some of the common observations we made were that users were able to find the camera button immediately; however, some of them went to the search bar instead. The camera page could be more user-friendly, and the upload button wasn't well understood. For older people, the cropping page was difficult to navigate, and a user stated that instead of cropping, they would simply retake the picture. Another user suggested that they would have liked to see the language selection page before scanning the menu. For more details of a dish, people either clicked on the name of the dish or didn't know that the menu items were even clickable. People stated that they would like to see the original menu rather than a formatted one. In the detailed dish page, the volume button went unnoticed, and the translation button was confusing. In the review page, most people would rate it with stars instead of giving written feedback.

After we discussed our observations, we decided to come up with less confusing icons at the bottom of the screen to make sure our users do not mistakenly go to another tab that's not the camera page. We will also consider redesigning the camera page so that it is easier for the user, especially the upload button. Instead of always taking the user to the cropping page, we might make it a separate button labeled "crop" so that the users can use

it only if they want to. We will also add a submit button to the review page and use IPA instead of a volume button. For the translation button, we will add a label underneath a newly designed icon, which will be easier to understand. We will also place the language selection page before the users scan the menu for a smoother experience and add questions to the review page instead of having a blank text box.

Hypothesis:

From the home page, the user would first notice the camera icon and search bar according to their task of translating a menu. The camera page looks like the camera from other photo taking apps, therefore, it should be easy to navigate through. Leading to the "adjust" photo page, some people could find it unnecessary or helpful depending on their situation because taking a bad picture, the user may also go back and retake. In regards to the language page, we think that the UI can be improved so that the screen looks fuller and more aesthetic. The (french) translated menu would be familiar to people who use takeout delivery apps, but may be difficult without categorical outlines. The detailed dish pages may be overloaded with information for users to take in all at once. The rating page in the end is straight forward but we realize that we lack a button for submission.

Observations:

Home page:

- Camera button and menu button gets confusing at times
- went straight to the search bar and assumed that any restaurant can be searched up
- went straight to camera icon x2
- went straight to searching the restaurant name or choosing the gypsy restaurant if at the restaurant

Camera page:

- Design for camera could have been better
- Upload button was not well understood
- Upload button wasn't clear until they read the heading/title
- Camera should have some tweaks to make it more user friendly

Crop page:

- Some people did not click confirm after cropping
- similar to traditional camera cropping; comments that older people might not know what to do
- would have retaken the picture if the original image was bad and had a lot of background

Language Selection Page

- would have liked the language selection page in the beginning
- longer list of languages might be useful, comparing google translate familiar a lot
- adding more languages

Translated Menu Page:

- People clicked the name of the dish for the details to show up
- would've liked using the original menu rather than a doordash formatted menu like google translate camera option, where words were translate
- didn't know the menu items was clickable
- google translate often miss translate words, so better translation is always better
- Translated Menu want an option to do foreign currency exchange in Euros than USD
- Expecting words to be translated like google translate

French Detailed Dish page:

- Translation button was confusing
- Volume button went unnoticed
- Confused by the "Done" button
- one interviewee commented that; for non-foreign users they wouldn't really use the app unless the menu dish had nutritional options such as protein, carbs, calories, since they workout a lot. Food description needs to be more intuitive. Consider Allegories and religious(Ingredient is very important)
- Confusing translation button
- noticed the speaking icon didn't understand why it was there since the person already speaks french
- would be nice if there was a speaking option to communicate with waiter, since google translate has that options to do voice

In the detailed dish page, the volume button went unnoticed and the translation button was confusing.

English Detailed Dish page:

- People questioned the necessity (x2)

- There should be a "save dish" option so that there can be a list of items to order generated for the user in the end (like a shopping cart)
- Skipped over most of the time since, it was similar to the previous page.

Review page:

- Stars went unnoticed
- People typed in the feedback interview said they wouldn't write a review rather they would just rate it with stars
- Make the reviewing page easier to do, since a lot of people don't want to think about what to write, so having suggesting questions or an incentive to review would be nice.
- Most would just rate it with a stars

Improvements:

- Try to find a new icon for the translation button + add a label
- Instead of taking the user to the cropping page every time, have a button labeled "crop" so that they can choose to crop or not
- Add a submit button to the rating page
- Brainstorm better design for camera?
- Move the volume button somewhere else so it's noticeable?
- Make the camera screen the "home" page and move the actual home page towards the end of the flow
- option to do foreign currency exchange in Euros than USD
- Compared with google translate, google translate is more confined with its ability to do restaurant ordering, our app might be better.
- having a voice memo icon to use to speak with a waiter
- have more
- Could I remove the translation button altogether?
 - French/Eng name instead of translating the entire screen
- Change volume button to IPA

We will try to come up with less confusing icons at the bottom of the screen to make sure our users do not mistakenly go to another tab that's not the camera page. We will also consider redesigning the camera page so that it is easier for the user, especially the upload button. Potentially, we could use the camera roll as an icon. We need to find a way for the cropping to be confirmed by the user, perhaps a signifier or some better design.

Think-Aloud Script

Think-Aloud Script

[[Home Page](#)]

- Tell me what you notice on this page. What would you click on first?

[[Camera Page](#)]

- What would you do here? What do you expect will happen after this?

[[Crop Page](#)]

- What would you do to make sure the app crops the right part of the menu?

[[Language Selection Page](#)]

- How would you select the language you want?

[[French Menu Page](#)]

- What do you notice about the translated Menu?

[[French Dish Page](#)]

- Does this translation give you enough information to understand the dish?

[[English Dish Page](#)]

- Would you want to be able to switch back and forth between the two languages?

[[Review Page](#)]

- How likely would you leave a review after the dish page?
- What would you write?

Think-Aloud testing

Think-Aloud Task

Imagine you are in the U.S. looking at an English menu, but you only speak French. Use the app to translate the menu into French

[[Task 1](#)] - *Find the option to Scan a Menu*

[[Task 2](#)] - *Take a photo of the menu*

[[Task 3](#)] - *Adjust the menu photo so only the menu is visible*

[[Task 4](#)] - *Select translation from English to French*

[[Task 5](#)] - *View the translated menu and tap a dish to view its details*

[[Task 6](#)] - *View the dish in French details*

[[Task 7](#)] - *Find the option to view the details of the dish in English*

[[Task 8](#)] - *Leave feedback on your experience*

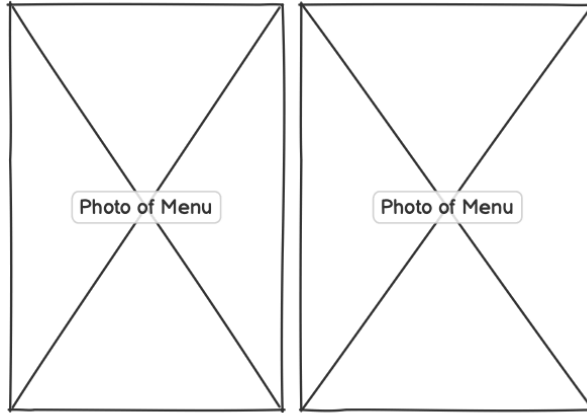
Home Page

09:52 AM



Recently Translated

Q search

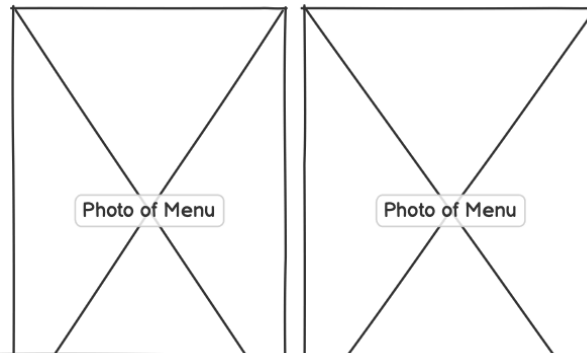


...

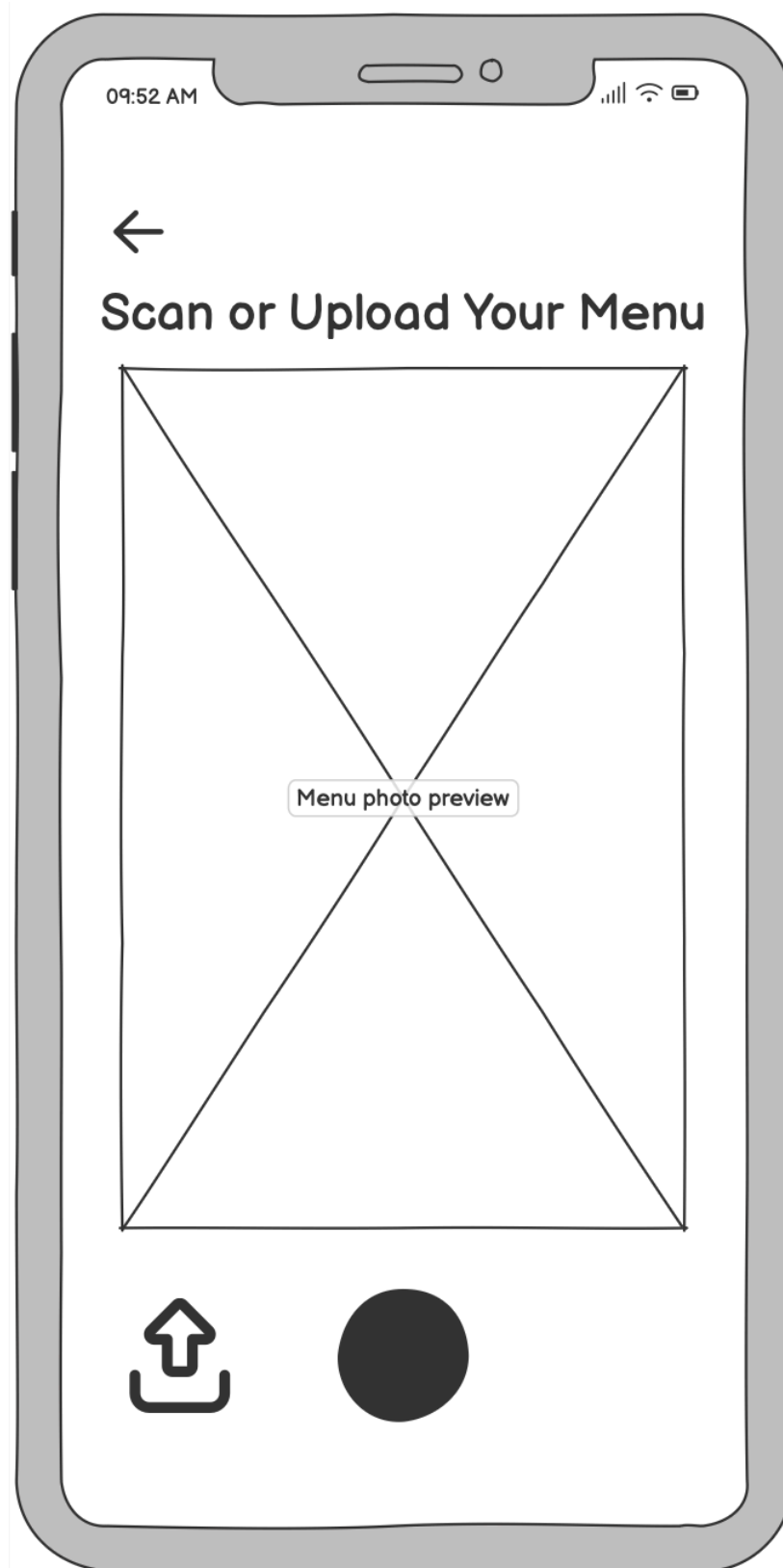
Gypsy's Trattoria Italiana



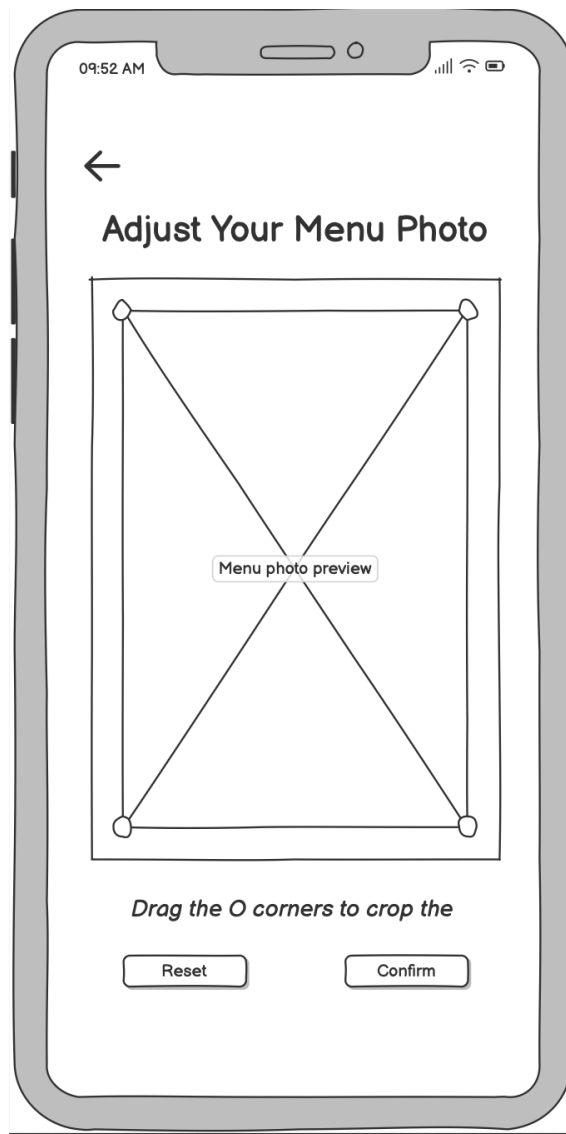
Berkeley, CA, USA
Italian Cuisine

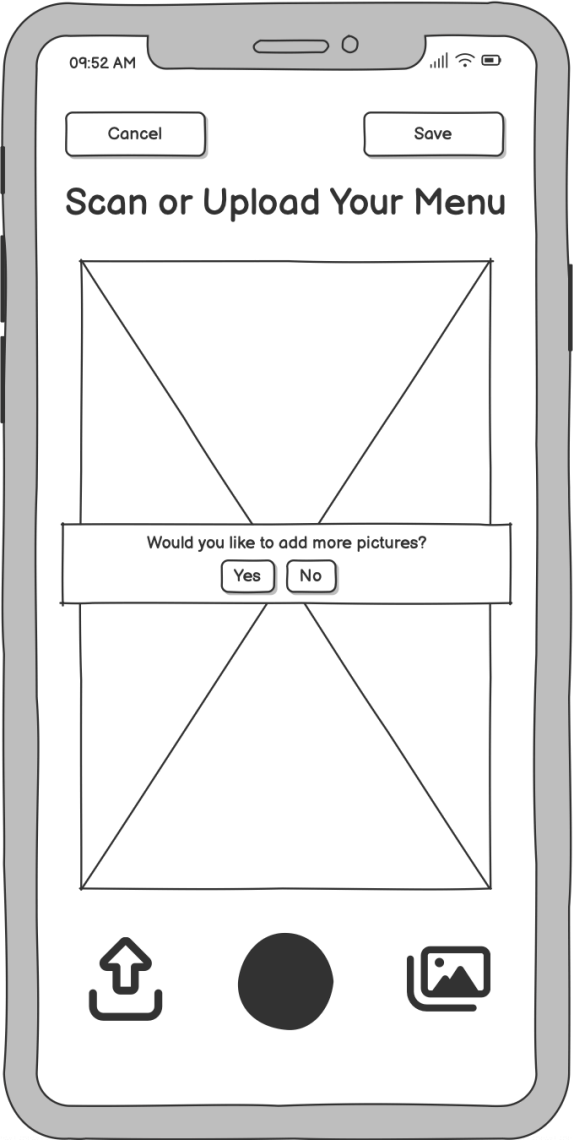


Camera Page

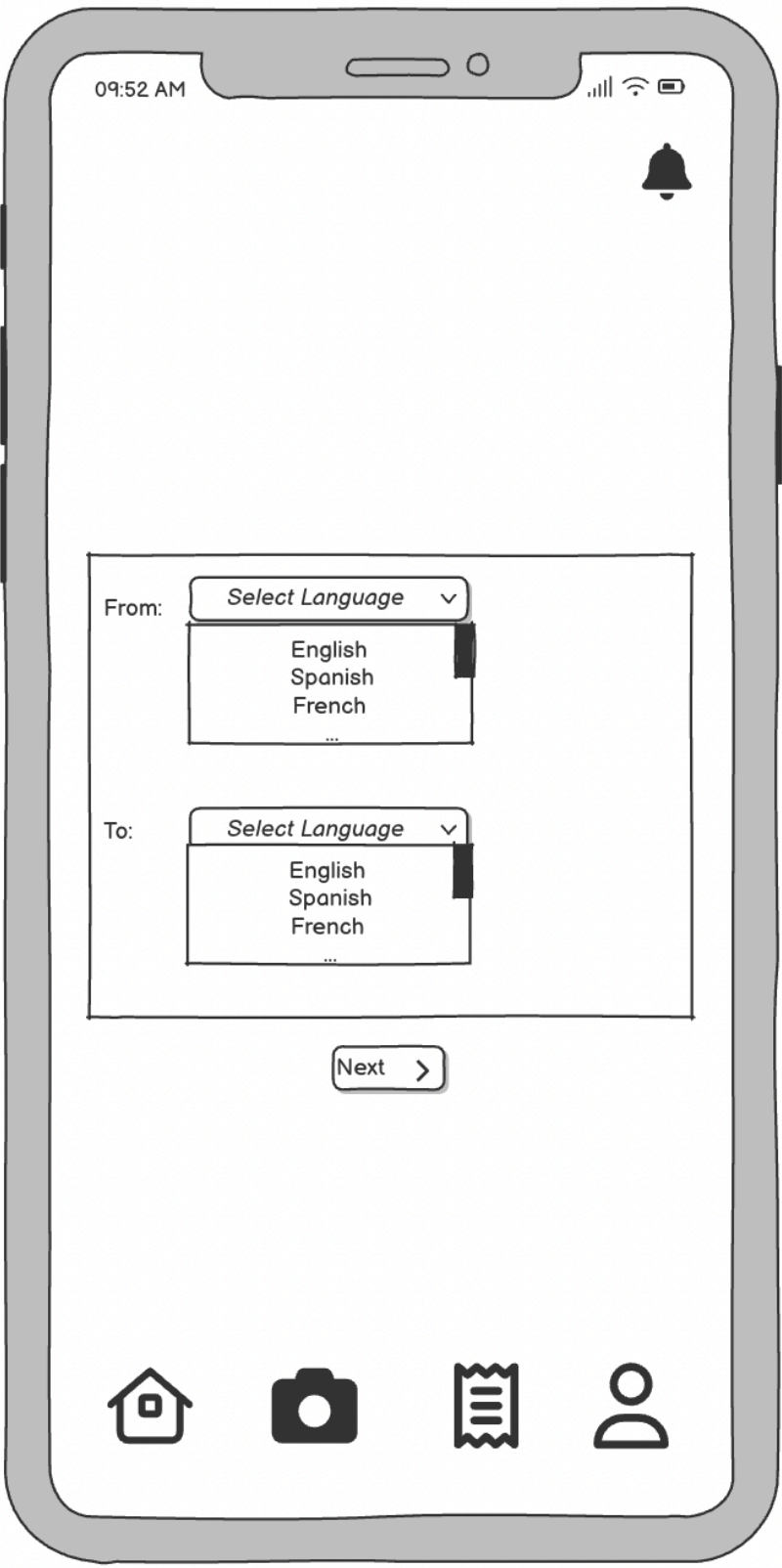


Crop Page





Language Selection Page



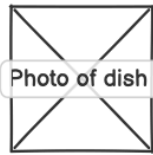
French Menu

09:52 AM



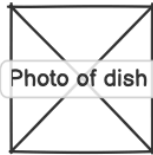
Your Translated Menu

Gypsy's Trattoria Italiana Menu



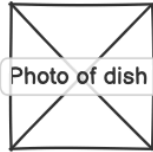
Poulet Marsala
plus d'infos...

\$14.99



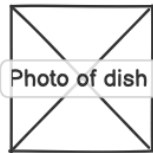
Gnocchis au poulet
plus d'infos...

\$14.99



Poulet gitan
plus d'infos...

\$14.99

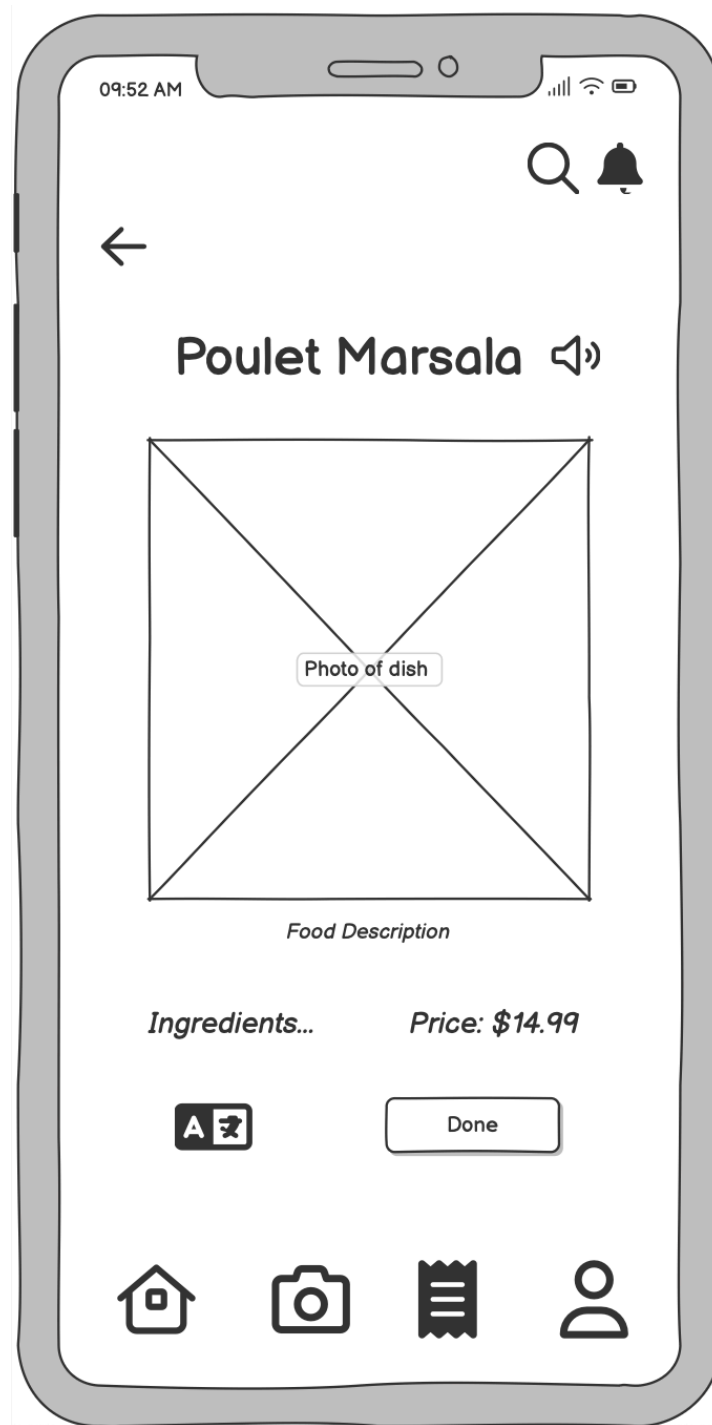


Poulet Parmigiano
plus d'infos...

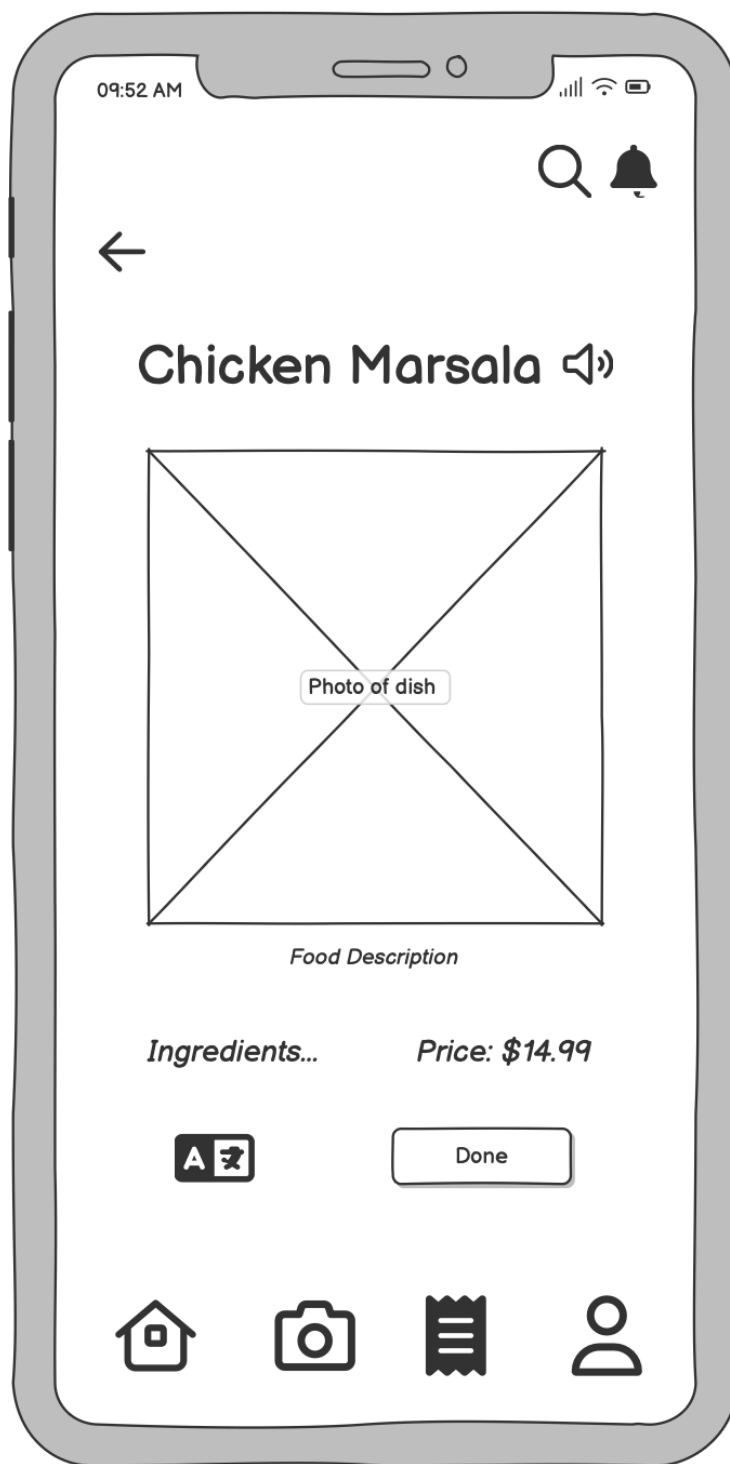
\$14.99



French Dish



English Dish



Review Page



Concept Test

After Interview Questions

(1-3 questions you want to ask the user about your app):

1. How would you describe your overall experience using this app?
 2. Were there anything unexpected/confusing after clicking on something?
 3. What did you like and dislike about previous translation apps you've used?
-

Eden's Interview Notes - user testing

EDEN's Interview Notes

Interview #1

Date: August 15, 2025

Time: 2:15PM

Interviewee Description: Male third year studying MCB, pre-Med age 19 from SoCal

Documented Questions & Answers:

what the user did	what the user said	observation/issue
clicking on the camera on the home page	camera is intuitive	camera icon
Take a photo of the menu	the uploading icon might be a little bit confusing if they don't read the words	no comments
cropped image like the cropping feature on the camera app	older person might not know what to do	no comments
Select translation from English to French	google translate selecting to and from better familiarity with google translate having a similar format might be better Translation page: can be improved maybe a little bit confusing, knowing what translate	no comments
View the translated menu and tap a dish to view its details	I didn't know we could click the menu items	more evidence to click on menu item for more information
View the dish in French details	Very overloaded on information. Food Description I would like nutrition such as calories, protein, etc. Since I work out a lot.	allergies and religious reason on the food ingredient might be useful
rated on stars, writing a review might take too	I wouldn't want to write anything since I want to	mostly review with the stars not bothered to

much time at the end.	get the food or is already eating. No incentive.	write a review, having questions for the review might motivate for a review
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How would you describe your overall experience using this app?

I won't really use this app, don't know if it will be useful, though it is easy to follow and go through, not confused with any tools. It might be useful going to a foreign country and not knowing the dishes.

Were there anything unexpected/confusing after clicking on something?

nothing was confusing, everything felt intuitive

What did you like and dislike about previous translation apps you've used?

Dislike: Google translate sometimes can't capture the translation that well and it doesn't identify the app that well.

Dislike: Better interface and options, google translate is confined compared to our app. would be very useful in foreign countries. Having nutrients displays such as a protein is more useful for more natural users not in a foreign country. Compared to google translate our app has less hassle as it combines a lot of stuff.

Interview #2

Date: August 10, 2025

Time: 3:02PM

Interviewee Description:

Documented Questions & Answers:

Female MCB and Business, 3rd year, 20 from Bay Area

Female 3rd year, 19, SoCal studying Microbial Biology

what the user did	what the user said	observation/issue
search restaurant if is available	there is gypsy restaurant options	camera icon, scan the menu. camera view and take a pictures
camera view and take a pictures	no comments	no comments
cropping picture with background	straight forward cropping nice words	pretty straight forwards, if the original image is bad it might not be good so I would retake it for better cropping.
from english to french	google translate expect the words to be translated like google translate longer list on language might be useful translating may be better before everything else such as scan the menu	might be better in the beginning with the language options
View the translated menu and tap a dish to view its details	photo name and current price	no comments
View the dish in French details	would like looking at the cost before and after, for example in USD and Euros	no comments

Find the option to view the details of the dish in English	speaking icon expected to be in english since you already speak french icon at the bottom might be a translate icon helpful can i have something with other menu item itself	no comments
Leave feedback on your experience	won't really leave a review	having a selection options on feedback such as a happy face, neutral face, or sad face

How would you describe your overall experience using this app?

- initiative easy to use, easy to follow along, person using this the first time would be able to follow along easily
- pressing the dish on ingredient will be very easy, dishes in another country might be confusing,

Were there anything unexpected/confusing after clicking on something?

- the currency might be very important

What did you like and dislike about previous translation apps you've used?

- all the ingredient
- google translate doesnt do currency switch
- google translate must h be the same with same image
- speaking on the conversion on google translate
- speaking with the waiter with google translate
- doing everything on the app
- Google might not be able to translate the words correctly.

Mihi's Interview Notes

[Mihi]'s Interview Notes

Interview #1

Date: August 10th

Time: 1:07 PM

Interviewee Description: 22-year-old who loves to try new cuisines

Documented Questions & Answers:

what the user did	what the user said	observation/issue
Clicked on the camera icon	I go here [to the dashboard] and click on the camera	- Quick to navigate the page and finish the task
Found the camera button	I don't like the camera, the design could be better	Understood how to click a picture but would have preferred another design for the camera
Dragged the o corners to fit the menu → clicked no → clicked on save	Ok, I drag these circles until i see only the menu and i don't want to add more pictures and I guess I hit save since there's only one menu. I could also click the photos app and see through my pictures to find the menu. The two screens were confusing	- Found it confusing, at first, didn't understand the reasoning behind the feature - Didn't click confirm - Made note of the multiple images icon, but not the upload
Selected english for "from" and french for "to" and clicked on next	I'm going from english to french and then next	- Quick to navigate the page and finish the task
Clicked on the name of a dish	Everything is translated into French. I suppose I click on this [the name of the dish] to view it	- Observed that the menu was in french
Viewed the dish and hit done	"The random icon is confusing"	- confused about what the translation icon was supposed to do/ what the icon represented. - Didn't mention the volume button

Clicked on ingredients to convert it back into english	"Why do I need to translate it back into English if I only speak French?"	- The translation button is confusing - Mistake the ingredients button for the translation instead
Type in my feedback and click on the stars to rate it	"There is no send button"	- Need to add a submit button somewhere on the page

How would you describe your overall experience using this app?

- It was like any standard app
- Most of things were standard, some features were confusing
 - The review page did not have a submit button
 - The scanning page had 2 screens
 - Didn't understand what the translation button was

Were there anything unexpected/confusing after clicking on something?

- Yes, all the bullet points mentioned above

(review page did not have a submit button, scanning page had 2 screens, didn't understand what the translation button was)

What did you like and dislike about previous translation apps you've used?

- ChatGpt voice is really nice because it just used voice and it would translate it
- Google translate, Google lens, or apple translate (one of these apps) has a feature where you can take a pictures and it translates the picture

Interview #2

Date: August 10th

Time: 1:37 PM

Interviewee Description: 75-year-old whose first language is Hindi

Documented Questions & Answers:

what the user did	what the user said	observation/issue
Clicked on the menu button	I think clicking on the menu button should take me to the scan it	- Instead of the camera button, they clicked on the menu icon instead

Clicked on the camera button	I think I should click here, I don't understand what the icon next to it means	- Found the camera button, however didn't understand what the upload icon meant
Click confirm and then yes until they were done adding images. Then they clicked no and hit save	"What do I do with the corners?"	- Spent the most time on these screens - It was difficult for them to navigate these 2 screens, especially the cropping page
Clicked English for "from" and "French" for "to" and hit next	So to go from one language to another, I should select English here and French here.	- This page was efficient
Saw the menu and clicked on the name of the first dish	To pick one dish, I guess I'll click on the first one	- Clicked on the name of the dish to enlarge it
Hit done	Made note of the photo and food description	Didn't look through the features that much
Didn't really understand the task/ where to find the icon to do so	I don't really understand what to do?	- The translation button was confusing
Typed in the feedback	I write my feedback in here	- Didn't notice the stars button

How would you describe your overall experience using this app?

- It was ok, I was confused a lot throughout though

Were there anything unexpected/confusing after clicking on something?

- The cropping page and English dish tasks were confusing
- Nothing unexpected, but it wasn't fast and efficient for the user either
 - The translation button
 - The cropping page

What did you like and dislike about previous translation apps you've used?

- Haven't used any translation apps before

Samia's Interview Notes

Samia's Interview Notes

Interview #1

Date: Aug 10, 2025

Time: 9:15AM

Interviewee Description: Family Member, likes to travel

Documented Questions & Answers:

what the user did	what the user said	observation/issue
• Uploaded a photo of a restaurant menu • Clicked Translated section • Tried to translate menu items themselves	• "I found it easy to upload the menu photo, but wasn't sure if I could crop it." • "I like seeing my history, but the search icon isn't very noticeable." • "Translations were fast, but some dish names didn't translate well."	• Search icon design not prominent. • Certain dish names poorly translated.

How would you describe your overall experience using this app?

I think the app is very useful for translating menus when I'm traveling. The interface is mostly straightforward. I wish the design was a bit more playful.

Were there anything unexpected/confusing after clicking on something?

When I clicked on the search button, I didn't realize that it was for searching past translations. At first, I thought it would search for restaurants.

What did you like and dislike about previous translation apps you've used?

I like when translation apps offer both text and photo. Some apps wouldn't let me save translations or review my history, which was inconvenient. I don't like when translations sound robotic or don't know names of local dishes.

Interview #2

Date: August 10

Time: 3:30PM

Interviewee Description: Berkeley student, undergraduate, likes to try new foods

Documented Questions & Answers:

what the user did	what the user said	observation/issue
• Opened app and tried scanning a takeout menu • Selected favorites tab • Changed language	• The camera scan was quick, but text on glossy paper was hard to capture • I like being able to save dishes, but it took me a while to find this • Changing language is simple, though some buttons stay in English	• Scanning works well, struggles with glare • Favorites feature useful • discoverability low • Language toggle intuitive, partial translation

How would you describe your overall experience using this app?

I enjoyed using the app for checking menu items before ordering. The layout is clear and it feels modern compared to older translation apps I've used

Were there anything unexpected/confusing after clicking on something?

When I clicked on 'Recently Translated,' I expected to see dishes sorted by date, but some older entries appeared at the top instead of my latest menu

What did you like and dislike about previous translation apps you've used?

When I clicked on 'Recently Translated,' I expected to see dishes sorted by date, but some older entries appeared at the top instead of my latest menu

Think-Aloud Script

[\[Home Page\]](#)

- Tell me what you notice on this page. What would you click on first?

[\[Camera Page\]](#)

- What would you do here? What do you expect will happen after this?

[\[Crop Page\]](#)

- What would you do to make sure the app crops the right part of the menu?

[\[Language Selection Page\]](#)

- How would you select the language you want?

[\[French Menu Page\]](#)

- What do you notice about the translated Menu?

[\[French Dish Page\]](#)

- Does this translation give you enough information to understand the dish?

[\[English Dish Page\]](#)

- Would you want to be able to switch back and forth between the two languages?

[\[Review Page\]](#)

- How likely would you leave a review after the dish page?
- What would you write?

Catherine's Interview Notes - user testing

[Catherine]'s Interview Notes

Interview #1

Date: Aug 10

Time: 1:30 pm

Interviewee Description: college student who loves food and is an active restaurant reviewer

Documented Questions & Answers:

what the user did	what the user said	observation/issue
Home page: - Search up the restaurant in the search bar - Click on the restaurant, either upload or choose a menu from the restaurant for translation	- did not understand the task	Icons and the layout may need to be altered for more clarity, maybe also have signifiers
Camera Page: - Took a photo	- I can take a photo or upload a photo - It would then load and be translated	Series of actions that the users have to take seem very smooth
Crop Page: drag the circular corners	- It is obvious that you can drag the circular corners - Maybe they can be highlighted into	Improvements can be made here to highlight the buttons

	another colors	
Language selection page: - Pressed English - Pressed French - Pressed next	- There should be a default language - Make this page fill up the screen a little better	UI needs to be improved, it looks significantly less aesthetic than other pages
French Menu Page: Taps the title	- The menu is in french - I can scroll through and see translations - I suggest having sections like appetizers, entrees, maybe even a search bar	An exhaustive list with no sections may be overwhelming
French Dish Page: Pressed the language swap button	- I am not sure which button to press when I want to return to the main menu - You wish that there is a translated version of dish - The sound may not work so well because the restaurant is a very busy and loud setting where it may be better for the waiter to read the menu item off the phone or the menu	This task did not require any action of pressing a button
English Dish Page: Tried to press "done" and "back"	- I would prefer to have the translation and the original dish	- The "done" button is confusing

	on the same page, i think this way is more user friendly, this feels like an overcomplication - Have saved items like adding to a shopping cart, have the app compile all the saved dishes	
Review Page Tapped the star rating and opted out in typing	If my experience was good, it would be because of the efficiency, ease of using the app and accuracy, so I would write something. I barely leave reviews personally and I would only leave a review unless I have strong feelings (negative or positive)	There was no button to submit the review

How would you describe your overall experience using this app?

I can see myself using this app in a foreign country, because google translate is confusing to read and inaccurate, and this app will help people with allergies from the ingredients

Were there anything unexpected/confusing after clicking on something?

It is mostly straight forward, some features like the audio were not discoverable and should be introduced in onboarding

What did you like and dislike about previous translation apps you've used?

I don't have too much experience with other translation apps other than google translate. I think that this menu app has its own niche

Interview #2

Date: Aug 10

Time: 3:10

Interviewee Description: minimalistic corporate intern who enjoys nice dinners after work

Documented Questions & Answers:

what the user did	what the user said	observation/issue
Did not press anything	did not understand the task	The task may have been confusing or the UI was confusing, user understood after I explained the features
Pressed the circular button	nothing	
Toggled the circles on the corners of the page, pressed confirm	This is pretty straightforward	User was able to complete the task
Pressed English, Pressed French, Pressed next	I pressed English, I pressed French, then I Pressed next	This step may involve too many buttons to press even though it is straightforward
Taps the title	I think it is natural to tapped here	The title should be made into a button
Pressed the language swap button	nothing	This task did not require any action of pressing a button
Pressed the language swap button	nothing	
Typed in the box, and then pressed the stars	There is nowhere to submit and I am lazy to write something down	There was no button to submit the review, maybe give a mcq instead of a free response

How would you describe your overall experience using this app?

It was smooth, pages were designed with clarity, visually minimalistic.

Were there anything unexpected/confusing after clicking on something?

Not really

What did you like and dislike about previous translation apps you've used?

For google translate, it is automatically applied to many websites but translations from it is non-native and word by word.

